



Family Tree as a Graph

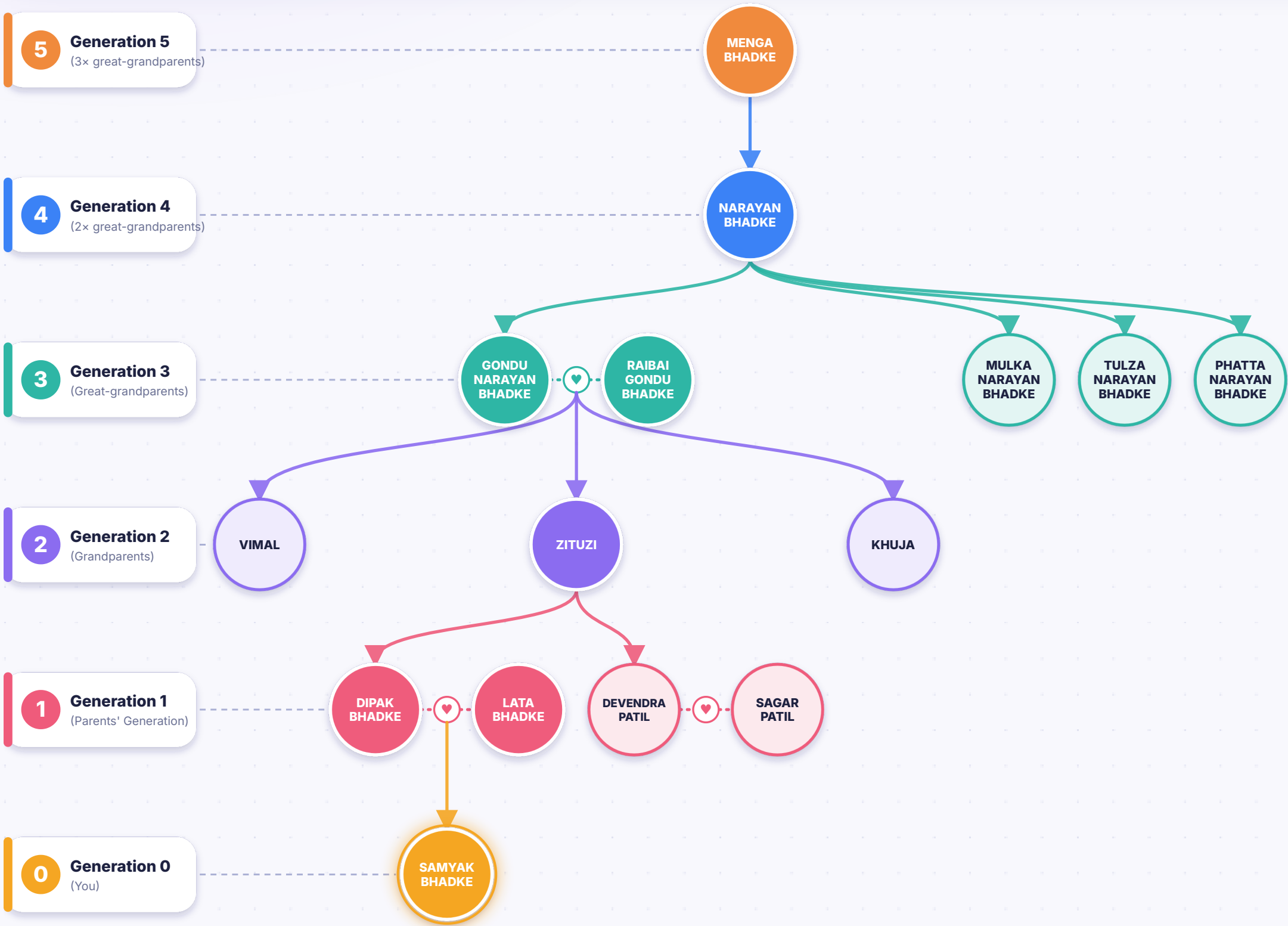
Discrete Mathematics & Graph Theory · The Bhadke Family

Family: **Bhadke**

Name: **Samyak Bhadke**

USN: **CS25041**

TAE-II DMGT



● You ● Direct ancestor (colour = generation) ○ Relative / spouse
---○--- Marriage → Parent – Child ----- Generation level

Family • Relationships • Graph Theory



Generation Gap from You (Samyak)

PERSON	RELATION TO YOU	GEN. GAP	EDGES
● Samyak Bhadke	You	0	0
● Dipak Bhadke	Father	1	1
● Lata Bhadke	Mother	1	2
● Devendra Patil	Aunt	1	3
● Sagar Patil	Uncle (by marriage)	1	4
● Vimal	Grandparent's sibling	2	4
● Zituzi	Grandparent	2	2
● Khuja	Grandparent's sibling	2	4
● Gondu Narayan Bhadke	Great-grandfather	3	3
● Raibai Gondu Bhadke	Great-grandmother	3	4
● Mulka Narayan Bhadke	Great-grandparent's sibling	3	5
● Tulza Narayan Bhadke	Great-grandparent's sibling	3	5
● Phatta Narayan Bhadke	Great-grandparent's sibling	3	5
● Narayan Bhadke	2× great-grandparent	4	4
● Menga Bhadke	3× great-grandparent	5	5



Shortest Path: You to Oldest Ancestor

Samyak → Dipak → Zituzi → Gondu → Narayan → Menga

5 edges / 5 generations



Key Observations

- Each person is a vertex (node): 15 vertices in this graph.
- Parent–child relationship is a directed edge (11); marriage is an undirected edge (3).
- Gen. gap = difference in generation level; Edges = length of the shortest path between two vertices.
- Without marriages: 12 vertices, 11 edges ($V - 1$) — connected and acyclic, so it is a tree.
- Narayan Bhadke has the most children here: 4.